Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of AI-based Autonomous Adaptive Welding Robot System Using Multi-Sensor for Harsh Optical Environments and Weld Joint Geometry Variations in Shipbuilding**			
Korea's 12 major national strategic technology fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (Major Category)**		**Technology Field 2 (Major Category)**	
	Mechanical·Materials		***Mechanical·Materials***	
	Medium category	Small category	Medium category	Small category
	Shipbuilding/Marine Systems	***Ship Production Systems/Construction Methods***	***Robot/Automation Machinery***	***Robot Control and Intelligence***
Proposed technology concept	**Core Concept** This technology aims to develop an 'Autonomous Adaptive Welding Robot System' that precisely recognizes the geometry of weld joints in real time, even under harsh optical conditions at shipbuilding welding sites, using multi-sensor. And an AI-based control algorithm optimizes welding parameters in response to changes in the work environment, such as thermal deformation. ● Development of an autonomous adaptive welding robot system for responding to harsh optical environments and variations in joint geometry at shipbuilding welding sites. In shipbuilding welding, ensuring visibility is challenging due to harsh optical conditions (intense arc flash, fumes, spatter, etc.). Consequently, arc sensing technology, which			

primarily utilizes changes in arc current and voltage, has been predominantly employed.

However, arc sensing detects only electrical signals rather than geometries, so it has limitations in addressing various geometric variables such as gaps and misalignment in welds caused by thermal deformation during welding.

Therefore, this project aims to establish a highly reliable AI-based monitoring system that is unaffected by harsh optical environments and geometric variables by combining electrical signals extracted via arc sensor with geometry information extracted through auxiliary sensors such as vision sensors.

Furthermore, the core objective is to establish an autonomous adaptive welding robot system where AI instantly calculates and corrects robot paths and welding parameters based on monitoring data to achieve consistent welding quality across various geometric variables.

Key Functions and Potential Applications

Key Functions

- Multi-sensor-based welding monitoring for harsh optical environments

  By Synchronizing the electrical signals (current/voltage waveforms) detected by the arc sensor with visual information or distance data acquired from auxiliary sensors such as vision sensors, unrecognizable areas are eliminated, and data reliability is maximized.

- AI-based real-time weld joint geometry recognition

  The AI algorithm removes noise from data collected via multi-sensor and infers the weld joint geometry in real time—including the molten pool boundary, gap variations in the welds, and misalignment—outputting pixel coordinate values of feature points and geometric metrics (gap values, misalignment angles, etc.) for use in robot coordinate transformation.

- AI-based real-time weld seam tracking

  Based on weld joint geometry information obtained through an AI model, the system infers the weld seam in real time. It continuously tracks the weld seam by correcting the robot path, accounting for sensor-processing-robot actuation delays.

- Adaptive control for autonomous optimization of welding parameters

  By utilizing geometry information of the recognized welds, the AI determines and calculates optimal welding parameters in real time, just like a skilled welder. It then transmits these parameters to the robot controller and welding machine to control heat input, thereby ensuring uniform weld quality and preventing welding defects.

Potential Applications

- Automation of thick plate welding processes in shipbuilding

  This solution can be applied to high-difficulty thick plate welding tasks that previously relied entirely on skilled workers' expertise, dramatically boosting productivity and serving as a core solution for building smart yards that address the deepening labor shortage problem.

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Rationale for Joint R&D Project	● Achieving high-reliability precision joining for shipbuilding and offshore structures When constructing high-value-added vessels and structures like LNG carriers and offshore plants, which demand durability in extreme environments and tolerate no defects, AI-based uniform weld quality control can be utilized as a highly reliable joining technology to ensure structural safety.
	Critical Technical Barriers ● Technical necessity Welding thick plates in shipbuilding involves high currents, generating an ultra-bright arc and large amounts of fumes. This makes it impossible to track the weld seam using ordinary cameras or existing laser sensors due to light scattering. Furthermore, thermal deformation during welding causes gap fluctuations and misalignment, making it difficult to infer the causal relationship between geometric variations and quality outcomes using a single sensor alone. In addressing these challenges, Korean SMEs possess strengths in productization and demonstration stages, including welding robot system integration, sensor protection design for field application, welding machine-robot controller linkage, and securing welding process data for thick plate welding in shipbuilding sites. However, to achieve performance levels suitable for on-site application—specifically, welding seam tracking and inferring causality between geometric variables and quality outcomes—requires integration beyond hardware system construction, encompassing advanced software and control technologies. Specifically, this necessitates processes such as converting large volumes of unlabeled sensor data into a learnable format, separating molten pool boundaries after noise removal in intense arc light environments, and optimizing AI models to simultaneously satisfy inference and control stability. However, developing these independently as an SME is expected to significantly increase development time due to the high development difficulty. ● Industrial necessity The shipbuilding industry is experiencing severe production disruptions due to a shortage of skilled welders relative to production demand. In particular, high-difficulty thick plate welding requires urgent automation, but it remains dependent on manual labor due to the harsh optical environment and the high technical difficulty posed by various geometric variables that occur during welding. Deploying unskilled workers frequently results in welding defects, incurring substantial rework costs. To resolve this, introducing intelligent robotic systems capable of replacing the judgment of skilled workers is essential. **Why Global Collaboration?** The autonomous adaptive welding robot system targeted by Korean SMEs is a highly challenging physical AI task that must simultaneously satisfy reliable data acquisition in harsh optical environments, real-time control stability, and adaptation to changing field conditions—going beyond simple hardware system integration.

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	To ensure the success of this project, it is essential to leverage Purdue University's world-class Physical AI core technologies, which are represented by research groups such as ICON (Institute for Control, Optimization, and Networks), XMO (eXcellence in Manufacturing and Operations), A2I2 (Assured Autonomy Innovation Institute), and LAMM (Laboratory for Multiscale Manufacturing). Specifically, Purdue has extensive experience in developing advanced perception-control integration algorithms and methodologies for field-level generalization, which are challenging for SMEs to develop independently. In particular, Purdue University's experience in conducting IoT-based Connectivity Projects using data obtained from local manufacturing companies, building real-time data integration frameworks, and performing research related to real-time data integration, cognition, and automation—such as Motion Digitalization using Vision Sensors—is expected to make a direct contribution to the successful execution of this project. Ultimately, by rigorously validating the system—integrated with Purdue's pioneering core technologies and Korean SMEs' manufacturing capabilities—in actual production environments, we expect to accelerate the development of a complete solution that can be immediately applied in real industrial settings, moving beyond the theoretical research stage.
Key words	1. Multi-Sensor based Welding Monitoring System 2. Autonomous Adaptive Welding Robot System 3. AI-based Weld Joint Geometry Recognition 4. AI-based Seam Tracking 5. Welding Parameter Optimization

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	**R&D Objectives** The objective of this project is to secure multi-sensor-based monitoring technology to overcome the harsh optical environment in shipbuilding welding sites, develop an AI-based autonomous adaptive welding robot system that tracks the weld seam in real time in response to geometric variables, and optimize process variables to prevent welding defects. **Detailed Objectives** ● Development of a multi-sensor-based monitoring module for harsh optical environments Development of a welding monitoring module integrating arc sensors with auxiliary sensors (e.g., vision sensors), capable of extracting geometry information of weld joints and molten pool even in environments with intense arc light and fumes. ● Development of AI-based welding monitoring algorithms Development of an AI model that recognizes gaps, misalignments, and molten pool boundaries in real time during welding by removing noise from geometry information extracted through multi-sensor. ● Development of an AI-based autonomous adaptive welding control algorithm			

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R&D Target Specifications

Development of a weld seam inference and tracking algorithm that corrects path errors in real time during welding.

Development of a control algorithm to calculate optimal welding parameters such as current, voltage, and feed rate for preventing welding defects.

- Establishment of an integrated robot system

Prototype development of an autonomous adaptive welding robot equipped with a developed multi-sensor module and AI-based control algorithm.

Evaluation of welding quality (ISO standard) and verification of system reliability through field testing in a similar industrial environment.

Key Performance Indicators	Unit	Final goal	Evaluation method
Processing speed	Hz	≥ 30	Cycle from frame acquisition in ROI to coordinate calculation
Seam Tracking Accuracy	mm	≤ ±0.5	Centerline deviation of weld seam
Gap Detection Range	mm	0.5 ~ 10	Gap Recognition Accuracy of AI Models
Welding Quality	Grade	Level C	ISO 5817 quality

R&D Plan

Key Tasks & Milestones

Year	Key Objectives	Major Tasks	Milestone
1	Construction of a Welding Dataset Based on a Multi-Sensor Module	- Development of a multi-sensor module to address harsh optical environments and geometric variables - Development of synchronization and preprocessing algorithms for electrical signal data with visual data or distance data - Construction of welding dataset under various conditions	- Multi-sensor module production - Achieved 90% recognition rate for molten pool boundaries
2	Control Algorithm Implementation and System Integration	- AI Model lightweighting and optimization for minimizing weld seam inference delays - Implementation of adaptive welding parameter control logic based on geometric variables - Prototype Development of Integrated Sensor-AI-Robot Controller	- Establishment of an Integrated Robot System - Ensure weld seam tracking error within ±0.5 mm
3	Performance Verification and Optimization	- Establishment and demonstration of a thick plate welding test-bed - Final system stabilization through welding quality assessment (ISO 5817) and feedback	- Achieved ISO quality grade Level C

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Schedule

Research Activity	1st year (Q3-Q4)		2nd year (Q1-Q4)				3rd year (Q1-Q4)				4th year (Q1-Q2)	
Design and Manufacturing of Multi-Sensor Modules												
Development of an AI Model for Geometry Recognition												
Development of Weld Seam Tracking and Adaptive Control Algorithms												
Performance Test Preparation and Verification												

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute

Purdue University's Role and Responsibilities

- Technical contributions & scope
  - Development of multi-sensor-based data fusion technology
    - Synchronization of the arc sensor's electrical signal and geometric information extracted from auxiliary sensors such as vision sensors.
    - Development of data preprocessing technology that extracts reliable data even in noisy welding environments.
  - Development of AI models for geometric shape recognition
    - Development of an AI model that analyzes multi-sensor-based data to infer geometry information such as gaps, misalignments, and root surfaces in real time.
- Specific tasks
  - Precise synchronization of heterogeneous data
    - Development of an alignment algorithm that synchronizes timestamps between data extracted from the arc sensor and auxiliary sensor within 10 milliseconds.
  - Preprocessing of data extracted through multi-sensor
    - Apply image pre-processing to suppress saturated arc-glare regions and perform contrast correction in fume-scattering regions.
    - Implement image filtering that extracts Gray-Level Co-occurrence Matrix (GLCM)-based texture features from vision-sensor images to enhance the molten pool boundary.

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	■ Data labeling using unsupervised learning-based AI algorithms · Transform large-scale data extracted via multi-sensor into latent space, perform unsupervised clustering in latent space to automatically cluster data patterns and generate pseudo-labels. · Use t-distributed Stochastic Neighbor Embedding (t-SNE) to visualize cluster separability and refine the labeling strategy. ■ Development of an AI-based real-time geometry recognition model · Extract a region of interest (ROI) from the welding images and implement and object-detection logic that identifies, at the pixel level, geometric feature points such as gap, and misalignment features within the ROI. · Optimize and lightweight the AI-based recognition model to minimize inference latency and meet the SME system requirement (image processing rate ≥ 30 Hz). ■ Model refinement and retraining based on field performance verification results · Based on the Korean SME's field-based validation results and failure-case logs, analyze root causes by error type, retrain the recognition model, and provide the SME with an updated model release.
R&R of Korean SME partners	Korean SME's Role and Responsibilities ● Developmental tasks ■ Development of specialized optical-based vision sensor modules · Development of a specialized optical module enabling observation of molten pool boundaries—difficult to identify with conventional cameras—under intense arc environments. ■ Implementation of high-speed precision motion control and weld seam tracking algorithms · Development of a path generation algorithm that converts geometry information such as the gap into the robot coordinate system for weld seam tracking and corrects the welding head position in real time. · Apply a predictive-control logic that compensates for latency between sensor measurements and robot motion to ensure tracking accuracy during high-speed welding. ■ Development of an algorithm for calculating optimal welding parameters · Development of an autonomous adaptive control algorithm that varies voltage, current, and feed rate based on geometric variables such as detected gaps and misalignment values to control heat input. · Improve the control algorithm by leveraging a data package in which in-process sensor data, robot control logs, and welding quality assessment results are linked under a common job ID. ■ Establishment of an integrated hardware system · Build an integrated hardware system that combines the developed multi-sensor module, an AI computing unit, and the robot controller.

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	● Testing & Validation ■ Establishment and operation of a test-bed · Production of a full-scale mock-up using AH36 and other high-strength plates employed in shipbuilding, replicating groove shapes and other features. · Intentionally vary the gap and misalignment conditions and conduct robustness tests of the developed autonomous adaptive welding robot system. ■ Welding quality assurance · Perform cross-sectional macrostructure analysis of the welds to verify compliance with the ISO 5817 quality level. ● Collaboration approach ■ Field data sharing and re-validation of improved AI models · Establish a continuous integration loop in which the Korean SME shares field-collected sensor and welding data with Purdue University, Purdue retrains and updates the AI models based on the field data, and the Korean SME deploys the updated models to the on-site autonomous adaptive welding robot system for re-validation.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	● Technological achievements ■ High-reliability multi-sensor fusion for harsh optical conditions and geometric variability · Combine Purdue University's AI models with the Korean SME's multi-sensor hardware platform and synchronize multi-sensor to reduce missing data and ensure consistent data acquisition. · Achieve seam tracking accuracy within ±0.5 mm even under high-current arc and fume conditions. ■ Implementation of an autonomous adaptive-control system embedding skilled welder expert decision-making · Develop an autonomous adaptive algorithm that goes beyond seam tracking by sensing gap variations and misalignment in real time and self-optimizing welding parameters accordingly. · Advance to TRL 5, suitable for real industrial environments from the lab-scale technology using AI models that guarantee uniform weld quality meeting ISO 5817 Level C or higher. ● Commercialization potential ■ Core solution for Smart Yard implementation · Apply the solution first to thick-plate welding processes in Korea's world-leading shipbuilding industry to significantly increase automation in operations that currently rely heavily on manual welding. · Conduct field tests with major Korean shipyards and their partner companies to establish reference cases and validate the reliability of the developed technology.

- Based on the autonomous adaptive welding robot system validated in Korea, expand deployment to the U.S. shipbuilding market, where demand for automated welding is expected to grow due to skilled labor shortages and increased production needs.

■ Strengthening bid competitiveness for high-value-added vessels and offshore structures

- Leverage high-integrity, defect-minimized welding capability as a key marketing differentiator to strengthen bid competitiveness in high-value-added manufacturing projects—such as LNG carriers and offshore plants—where welding defects can lead to catastrophic failures.

■ Horizontal expansion of the technology to heavy industry and construction machinery sectors

- Expand the technology's deployment to manufacturing lines for products requiring thick-steel welding—such as excavators, cranes, and wind-turbine tower substructures—to diversify target markets and maximize profitability.

● Industrial & Economic impact

■ Mitigation of manufacturing workforce decline and skilled-welder shortages

- Deploy robotic systems that can replace or assist skilled welders to improve production stability in the shipbuilding and heavy industries, where labor shortages are creating production bottlenecks.

■ Cost reduction through improved welding quality

- Use real-time adaptive control to prevent weld defects in advance, thereby reducing rework costs that represent a substantial share of overall process costs.

■ Early leadership in the global robotics market through strengthened U.S.–Korea technology alliance

- Establish a successful international joint R&D model by combining Purdue University's AI innovation with the Korean SME's manufacturing-grade hardware reliability.

6. References

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